

Data Analysis & Visualization using Power BI

Student Name: Adithyan R

Course: Data Analytics

Week: 5

Assignment Tasks

Task 1: Data Import / Loading

Task 2: Data Cleaning & Preparation

Task 3: Data Modeling & Relationships

Task 4: DAX Calculated Columns & Measures

Task 5: Power BI Visualizations

Task 6: Interactive Dashboard & Filters

Task 7: Business Insights & Recommendations

Task 1: Power BI Introduction & Data Import

1. Exploring the Power BI Interface

I explored the Power BI Desktop interface and its main features used for data analysis and visualization.

2. Importing the Dataset

I imported a sales dataset into Power BI Desktop for analysis.

3. Viewing the Imported Data

I viewed the imported dataset in Power BI to understand the available records and information.

4. Available Tables and Columns

The dataset contains the **Sales** table with the following columns:

- Date
- Product
- Category
- Sales

- Quantity
- Region

5. Data Types of Columns

Column	Data Type
Date	Date
Product	Text
Category	Text
Sales	Whole Number
Quantity	Whole Number
Region	Text

6. Purpose of Power BI in Data Analytics

Power BI is used to import, analyze, and visualize data. It helps users create reports and dashboards and understand data through charts, tables, and other visualizations.

Name	Date	Format	*Wednesday, Marc...	Summarization	Don't summarize
Data type	Date	Format	\$ % , .00 Auto	Data category	Uncategorized
Cancel	Structure	Formatting	Properties		
Date	Product	Category	Sales	Quantity	Region
Monday, January 5, 2026	Laptop	Electronics	60000	2	Kochi
Monday, January 12, 2026	Mouse	Accessories	2500	10	Kozhikode
Sunday, January 18, 2026	Keyboard	Accessories	4500	6	Kannur
Tuesday, February 3, 2026	Monitor	Electronics	18000	3	Kochi
Sunday, February 15, 2026	Laptop	Electronics	65000	2	Thrissur
Wednesday, February 25, 2026	Chair	Furniture	12000	4	Kozhikode
Wednesday, March 4, 2026	Headset	Accessories	3500	7	Kochi
Tuesday, March 17, 2026	Monitor	Electronics	20000	3	Kannur
Saturday, March 28, 2026	Keyboard	Accessories	5000	8	Thrissur
Monday, April 6, 2026	Laptop	Electronics	70000	2	Kochi

Task 2: Data Cleaning & Preparation

Task 2 PDF-il **ithu pole ezhuthuka:**

1. Inspecting the Imported Dataset

The imported sales dataset was inspected using **Power Query** to understand the data and identify any issues.

2. Identifying Missing or Incorrect Values

The dataset was checked for missing and incorrect values. No major missing values were identified in the dataset.

3. Removing Duplicate Records

The dataset was checked for duplicate records and unnecessary duplicate entries were removed where required.

4. Changing Incorrect Data Types

The data types of the columns were checked and corrected where required to ensure that the data could be used properly for analysis.

5. Renaming Columns

The column names were reviewed and renamed where appropriate to make them clear and understandable.

6. Removing Unnecessary Columns

Unnecessary columns, if any, were removed so that only relevant information was retained for analysis.

7. Applying Transformations

Suitable Power Query transformations were applied to prepare the dataset for analysis.

Task 3: Data Modeling & Relationships

Task 3 PDF-il **ithu pole ezhuthuka:**

1. Identifying Common Columns

The common column between the tables was identified as **Region**.

2. Creating Relationships

A relationship was created between **Sales[Region]** and **Table (3)[Region]**.

3. Relationship Type

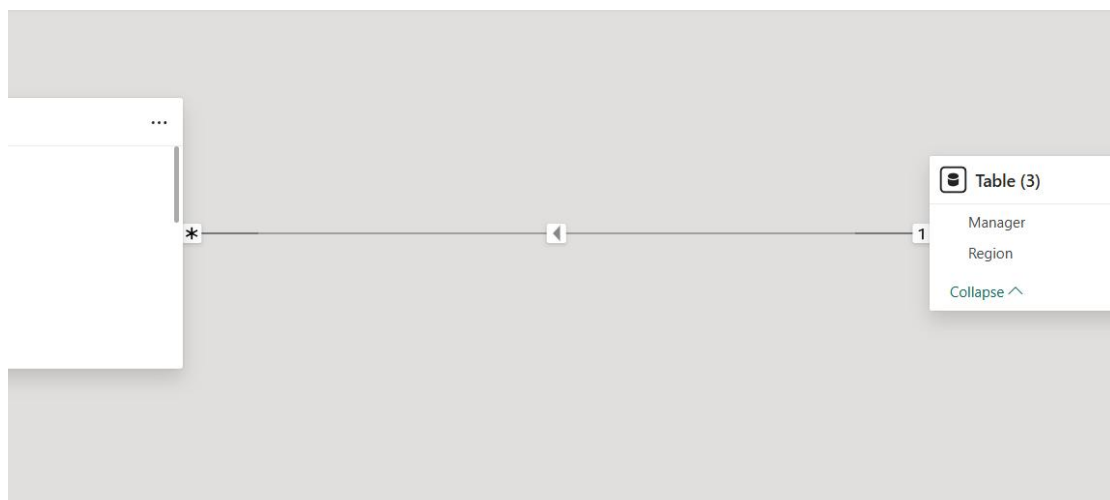
The relationship is a **Many-to-One (*:1)** relationship, where multiple records in the Sales table are connected to one corresponding region in Table (3).

4. Data Model

A basic data model was created by connecting the **Sales** table with **Table (3)** using the Region column.

5. Explanation of the Relationship

The relationship allows the Region information in **Table (3)** to be associated with the corresponding sales records in the **Sales** table. This helps in filtering and analyzing sales data based on region.



Task 4: DAX Calculated Columns & Measures

PDF-il **Task 4** section-il ithu pole ezhuthuka:

1. Total Sales

DAX Formula:

```
Total Sales = SUM(Sales[Sales])
```

Result: 260500

2. Average Sales

DAX Formula:

Average Sales = AVERAGE(Sales[Sales])

Result: 26050

3. Total Number of Records

DAX Formula:

Total Records = COUNTROWS(Sales)

Result: 10

4. Maximum Sales

DAX Formula:

Maximum Sales = MAX(Sales[Sales])

Result: 70000

5. Minimum Sales

DAX Formula:

Minimum Sales = MIN(Sales[Sales])

Result: 2500

6. Sales Percentage

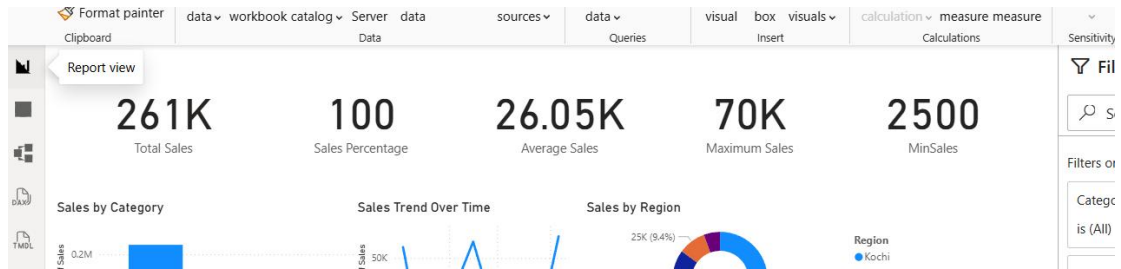
DAX Formula:

Sales Percentage = DIVIDE([Total Sales], [Total Sales], 0) * 100

Result: 100%

DAX Measures Created

At least 3 DAX measures were created and their results were displayed using **Card visuals** in Power BI.



Task 5: Power BI Visualizations

PDF-il **Task 5** section-il ithu pole ezhuthuka:

1. Bar / Column Chart

A **Column Chart** was created to compare sales across different product categories.

Title: Sales by Category

Axis: Category

Values: Sales

2. Line Chart

A **Line Chart** was created to show the sales trend over time.

Title: Sales Trend Over Time

Axis: Date

Values: Sales

3. Pie / Donut Chart

A **Donut Chart** was created to show the distribution of sales across different regions.

Title: Sales by Region

Legend: Region

Values: Sales

4. Table / Matrix

A **Table** was created to display the sales details clearly.

Fields used:

- Date
- Product
- Category
- Sales
- Quantity
- Region

Title: Sales Details

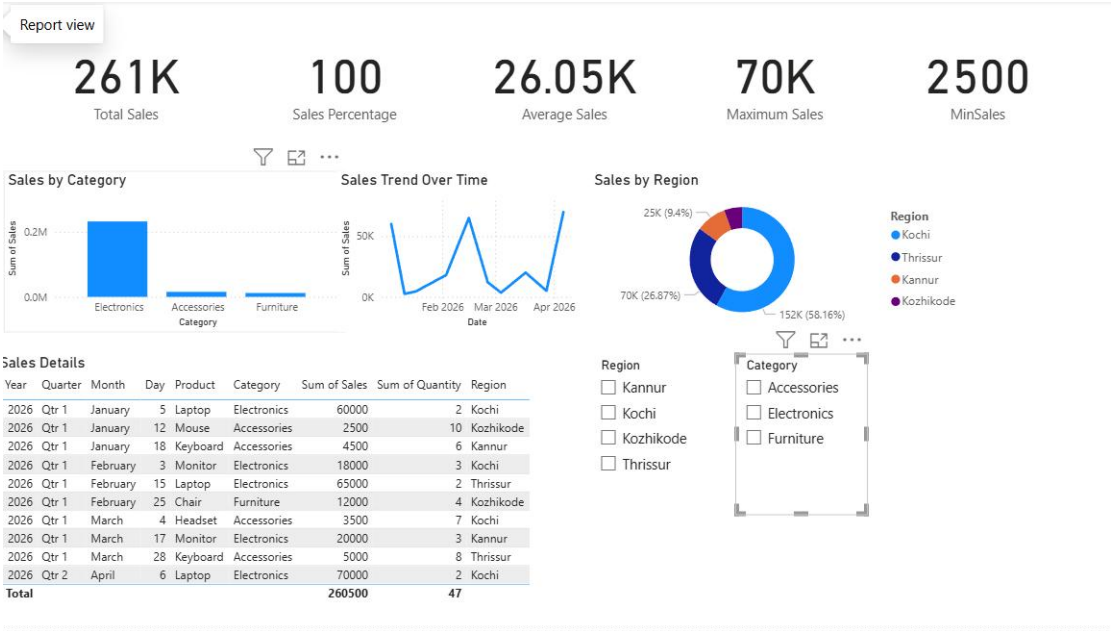
5. KPI / Card Visuals

KPI Cards were created to display important sales measures:

- Total Sales
- Average Sales
- Maximum Sales
- MinSales
- Sales Percentage

Result

Different Power BI visualizations were created to represent the sales data clearly and make the data easier to understand and analyze.



Task 6: Interactive Dashboard & Filters

PDF-il Task 6 section-il ithu pole ezhuthuka:

1. KPI Cards

At least three KPI Cards were created to display important sales metrics such as:

- Total Sales
- Average Sales
- Maximum Sales

- Minimum Sales
- Sales Percentage

2. Different Visualizations

The dashboard contains different types of visualizations to represent the sales data clearly:

- Column Chart
- Line Chart
- Donut Chart
- Table

3. Interactive Slicers / Filters

Interactive slicers were added to allow users to filter and explore the dashboard.

Slicers used:

- Region
- Category

4. Dashboard Layout

The visualizations, KPI cards, table, and slicers were arranged properly on a single dashboard page for easy understanding and analysis.

5. Interactivity

The slicers allow users to select a particular **Region** or **Category**, and the dashboard visuals update according to the selected filter.

Result

An interactive Power BI dashboard was created with KPI cards, multiple visualizations, and slicers for exploring the sales data.

Task 7: Business Insights & Recommendations

PDF-il **Task 7** section-il ithu pole ezhuthuka:

1. Business Insights

Insight 1:

Electronics is the highest-performing category based on total sales.

Insight 2:
Accessories and Furniture have lower sales compared with Electronics.

Insight 3:
Kochi is the highest-performing region based on total sales.

Insight 4:
Kozhikode is the lowest-performing region based on total sales.

Insight 5:
Sales vary over time, with the highest sales value reaching ₹70,000.

2. Business Recommendations

Recommendation 1:
Focus more on Electronics products through better stock availability and promotional activities, as Electronics generates the highest sales.

Recommendation 2:
Use targeted offers and marketing activities in lower-performing regions such as Kozhikode to improve sales performance.

Conclusion

The dashboard shows that **Electronics and Kochi are the strongest performers**, while **Furniture and Kozhikode have lower sales**. These insights can help the business focus on high-performing areas and improve weaker areas.

